

Customer Journey & Conversion Analytics

Dataset Overview

Rows: 12,330

Columns: 18

FOR A/B testing we will take the parameter 'Weekend'

Weekend: 0 → weekday || Weekend: 1 → weekend

Phase 1: Business Understanding

- What drives purchases?
- Which traffic sources convert best?
- Which months perform best?
- Does weekend traffic convert better?
- Which visitor types generate the highest revenue?
- How does engagement affect conversions?
- Can we predict whether a visitor will purchase?

Phase 2: SQL Project

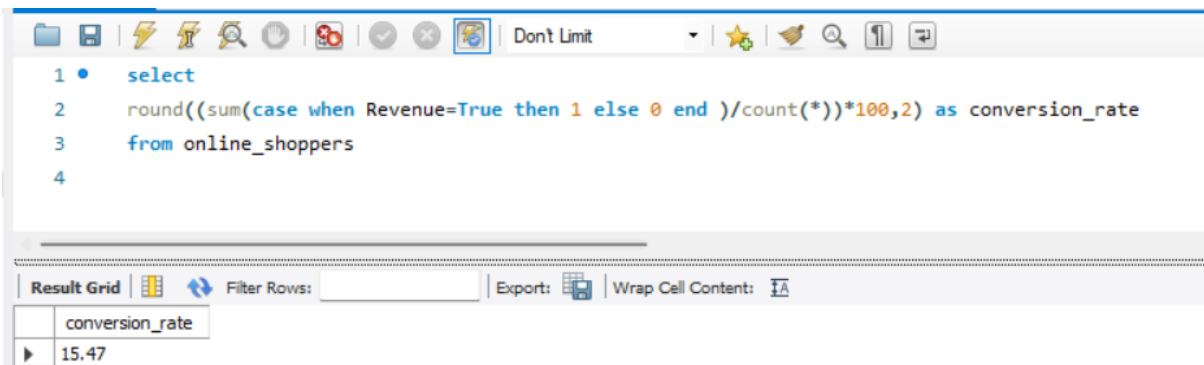
Create a SQL database

 `customer_journey_and_conversion_analytics`

Creating the Table

```
SQL File 7* x SQL File 5* SQL File 6*
create table online_shoppers (
  Administrative int,
  Administrative_Duration float,
  Informational int,
  Informational_Duration float,
  ProductRelated float,
  ProductRelated_Duration float,
  BounceRates float,
  ExitRates float,
  PageValues float,
  SpecialDay float,
  Month VARCHAR(10),
  OperatingSystems int,
  Browser int,
  Region int,
  TrafficType int,
  VisitorType varchar(50),
  Weekend boolean,
  Revenue boolean
);
```

1. Overall Conversion Rate

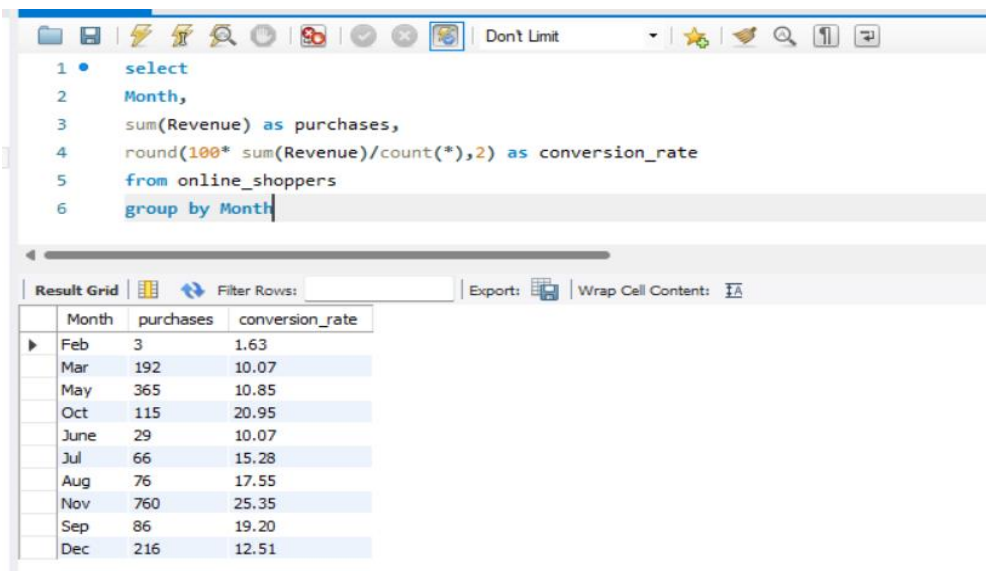


```
1 • select
2   round((sum(case when Revenue=True then 1 else 0 end )/count(*)*100,2) as conversion_rate
3   from online_shoppers
4
```

Result Grid

conversion_rate
15.47

2. Conversion by Month

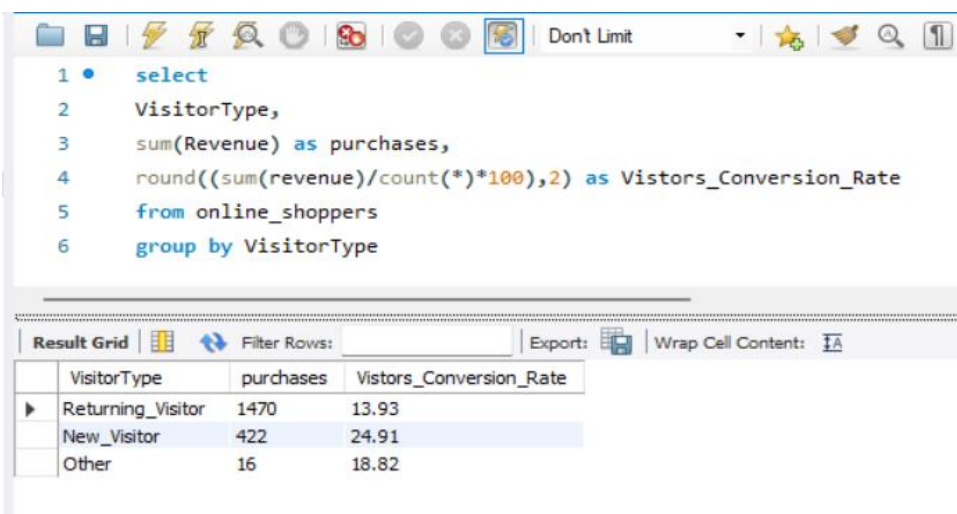


```
1 • select
2   Month,
3   sum(Revenue) as purchases,
4   round(100* sum(Revenue)/count(*),2) as conversion_rate
5   from online_shoppers
6   group by Month
```

Result Grid

Month	purchases	conversion_rate
Feb	3	1.63
Mar	192	10.07
May	365	10.85
Oct	115	20.95
June	29	10.07
Jul	66	15.28
Aug	76	17.55
Nov	760	25.35
Sep	86	19.20
Dec	216	12.51

3. Conversion by Visitor Type

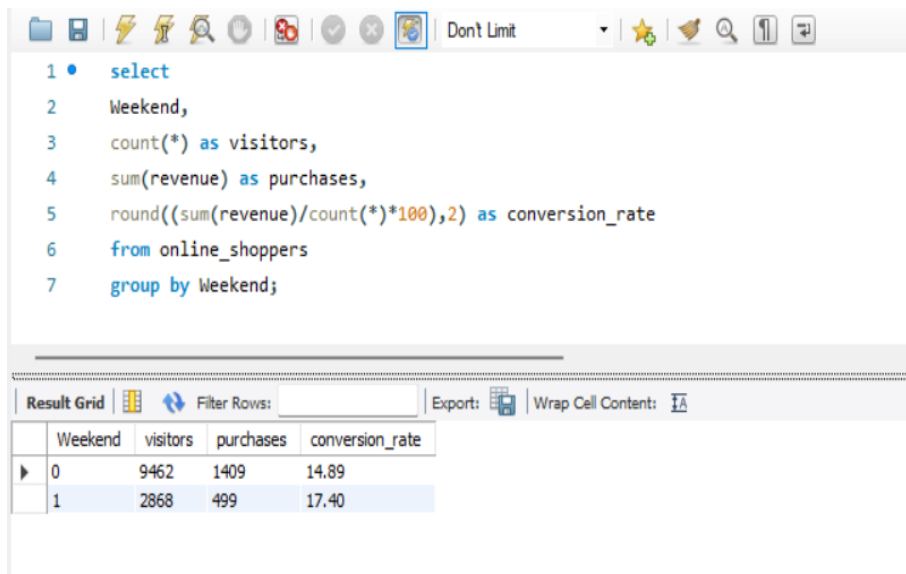


```
1 • select
2   VisitorType,
3   sum(Revenue) as purchases,
4   round((sum(revenue)/count(*)*100),2) as Vistors_Conversion_Rate
5   from online_shoppers
6   group by VisitorType
```

Result Grid

VisitorType	purchases	Vistors_Conversion_Rate
Returning_Visitor	1470	13.93
New_Visitor	422	24.91
Other	16	18.82

4. Weekend vs Weekday



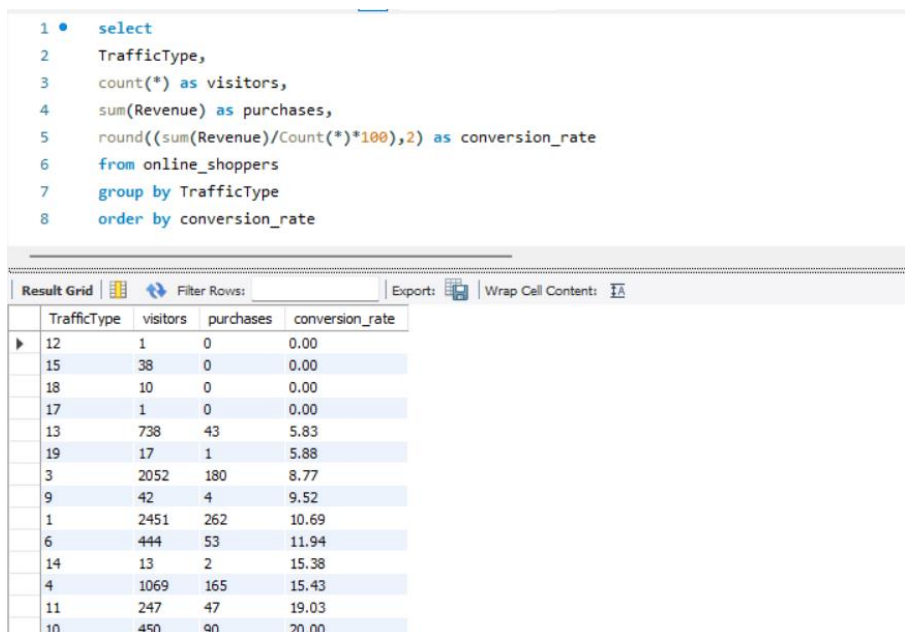
The screenshot shows a SQL query editor with a toolbar at the top. The query is as follows:

```
1 • select
2   Weekend,
3   count(*) as visitors,
4   sum(revenue) as purchases,
5   round((sum(revenue)/count(*)*100),2) as conversion_rate
6   from online_shoppers
7   group by Weekend;
```

Below the query editor is a results grid with the following data:

	Weekend	visitors	purchases	conversion_rate
▶	0	9462	1409	14.89
	1	2868	499	17.40

5. Best Traffic Sources



The screenshot shows a SQL query editor with a toolbar at the top. The query is as follows:

```
1 • select
2   TrafficType,
3   count(*) as visitors,
4   sum(Revenue) as purchases,
5   round((sum(Revenue)/Count(*)*100),2) as conversion_rate
6   from online_shoppers
7   group by TrafficType
8   order by conversion_rate
```

Below the query editor is a results grid with the following data:

	TrafficType	visitors	purchases	conversion_rate
▶	12	1	0	0.00
	15	38	0	0.00
	18	10	0	0.00
	17	1	0	0.00
	13	738	43	5.83
	19	17	1	5.88
	3	2052	180	8.77
	9	42	4	9.52
	1	2451	262	10.69
	6	444	53	11.94
	14	13	2	15.38
	4	1069	165	15.43
	11	247	47	19.03
	10	450	90	20.00

6. Monthly Revenue Contribution

```
1 • select
2   Month,
3   sum(case when Revenue = True then 1 else 0 end) as total_purchases,
4   round(
5     100*sum(case when Revenue=True then 1 else 0 end )/ count(*),2)as conversion_rate
6   from online_shoppers
7   Group by Month
8   order by conversion_rate desc
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

Month	total_purchases	conversion_rate
Nov	760	25.35
Oct	115	20.95
Sep	86	19.20
Aug	76	17.55
Jul	66	15.28
Dec	216	12.51
May	365	10.85
Mar	192	10.07
June	29	10.07
Feb	3	1.63

7. Returning vs New Visitor Performance

```
1 • select
2   VisitorType,
3   count(*) as visitors,
4   sum(Revenue) as purchases,
5   Round(
6     100*(sum(Revenue) / count(*)),2
7   ) as conversion_rate
8   from online_shoppers
```

Result Grid | Filter Rows: | Export: | Wrap Cell

VisitorType	visitors	purchases	conversion_rate
Returning_Visitor	10551	1470	13.93
Other	85	16	18.82
New_Visitor	1694	422	24.91

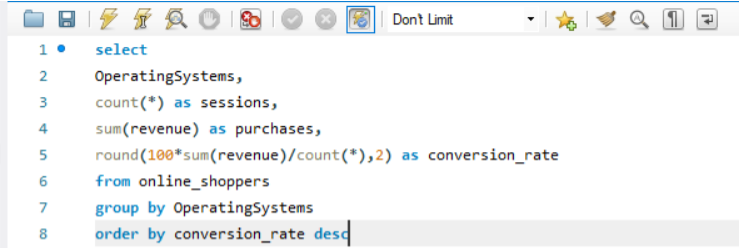
8. Which Browser Converts Best?

```
1 • select
2   Browser,
3   count(*) as sessions,
4   sum(revenue) as purchahses,
5   round(100*sum(revenue)/count(*),2) as conversion_rate
6   from online_shoppers
7   group by Browser
8   order by conversion_rate desc
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

Browser	sessions	purchahses	conversion_rate
12	10	3	30.00
13	61	16	26.23
10	163	32	19.63
5	467	86	18.42
4	736	130	17.66
11	6	1	16.67
8	135	21	15.56
2	7961	1223	15.36
1	2462	365	14.83
7	49	6	12.24
6	174	20	11.49
3	105	5	4.76
9	1	0	0.00

9. Operating System Analysis



```
1 • select
2   OperatingSystems,
3   count(*) as sessions,
4   sum(revenue) as purchases,
5   round(100*sum(revenue)/count(*),2) as conversion_rate
6 from online_shoppers
7 group by OperatingSystems
8 order by conversion_rate desc
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	OperatingSystems	sessions	purchases	conversion_rate
▶	8	79	17	21.52
	4	478	85	17.78
	2	6601	1155	17.50
	5	6	1	16.67
	1	2585	379	14.66
	7	7	1	14.29
	6	19	2	10.53
	3	2555	268	10.49

10. Region Wise Conversion Rate



```
1 • select
2   Region,
3   count(*) as sessions,
4   sum(Revenue) as purchases,
5   round(100*sum(Revenue)/count(*),2) as conversion_rate
6 from online_shoppers
7 group by Region
8 order by conversion_rate desc
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Region	sessions	purchases	conversion_rate
▶	9	511	86	16.83
	2	1136	188	16.55
	5	318	52	16.35
	1	4780	771	16.13
	7	761	119	15.64
	4	1182	175	14.81
	3	2403	349	14.52
	6	805	112	13.91
	8	434	56	12.90

➔ Important Metrics in this Dataset

- Product Related
- Product Related Duration
- Bounce Rate
- Exit Rate
- Page Value

11.Product Related

➔ Number of Product Related Pages Visited during the Sessions

⇒ How many product related pages did the user view

Do users who view more products convert better? ➔ Yes

```
1 • select
2   Revenue,
3   Round(avg(Productrelated),2) as avg_Product_Pages
4   from online_shoppers
5   Group by Revenue
```

	Revenue	avg_Product_related
▶	0	28.71
	1	48.21

Higher Value means:

- ⇒ Strong Engagement
- ⇒ Product Comparisons
- ⇒ Research Before Buying
- ⇒ Higher Purchase Intent

12.Product Related Duration

➔ Total Time in second that the users spend on a Product related page

```
1 • select
2   Revenue,
3   Round(avg(ProductRelated_Duration),2) as avg_product_duration
4   from online_shoppers
5   group by Revenue
6
```

	Revenue	avg_product_duration
▶	0	1069.99
	1	1876.21

⇒ Higher value means:

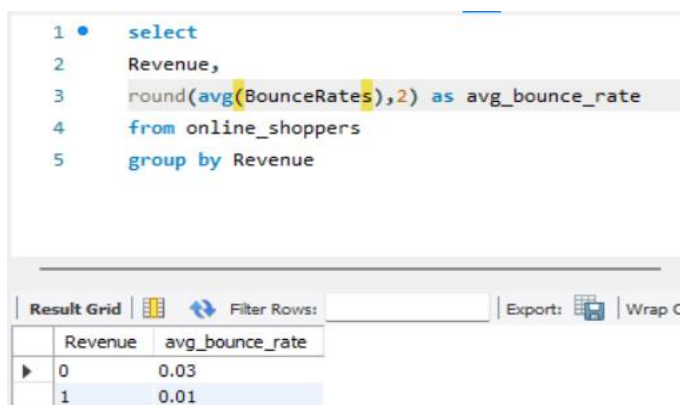
- Strong Purchase intention
- User researching Product
- User comparing option

⇒ Lower value means:

- Low Engagement
- Visitor not interested

13. Bounce Rate

➔ The Percentage of session where the user enters and leaves without meaningful interaction



```
1 • select
2   Revenue,
3   round(avg(BounceRates),2) as avg_bounce_rate
4 from online_shoppers
5 group by Revenue
```

	Revenue	avg_bounce_rate
▶	0	0.03
	1	0.01

Values: 0.01 = 1%

0.20=20%

0.50=50%

Higher Bounce Rate

1. Bad Landing Page
2. Irrelevant Traffic
3. Poor User Experience

Lower Bounce Rate

Engaged user
Better Quality Traffic
Good User Experience

Purchaser ➔ 0.01 = 1% ➔ Purchaser Explored.

Non-Purchaser ➔ 0.03 = 3% ➔ Non-Purchaser Left Quickly.

14.Exit Rates

➔ The percentage of page views that ended with users leaving the website from that page.

```
1 • select
2   Revenue,
3   round(avg(ExitRates),2) as avg_Exit_Rates
4   from online_shoppers
5   group by Revenue
```

	Revenue	avg_Exit_Rates
▶	0	0.05
	1	0.02

Higher Exit Rate

1. Confusing Page
2. Price Shock
3. Check-out Issue
4. Lack of Information

Low Exit Rate

User continue Deeper into funnel

15.Page Value

➔ A Score representing how valuable that page is.

⇒ Pages closure to conversion receives a higher value

```
1 • select
2   Revenue,
3   round(avg(PageValues),2) as avg_page_values
4   from online_shoppers
5   group by Revenue
```

	Revenue	avg_page_values
▶	0	1.98
	1	27.26

Higher Page Value

1. Strong Buying intent
2. User Closure to Conversion

Low Page Value

- Casual Browsing
- Low Commercial intent

Customer Segmentation

16. Engagement Segmentation

```
1 • select
2   case when ProductRelated < 20 then 'Low'
3     when ProductRelated < 50 then 'Medium'
4     else 'High' end as Engagement_Level,
5   count(*) as Users,
6   round(sum(revenue)/count(*)*100,2) as conversion_rate
7   from online_shoppers
8   group by Engagement_Level
```

Engagement_Level	Users	conversion_rate
Low	6508	9.83
Medium	3606	19.61
High	2216	25.32

17. Session Segmentation

```
2   case when ProductRelated_Duration < 500 then 'Low'
3     WHEN ProductRelated_Duration < 1500 then 'Medium'
4     else 'High' end as Session_Type,
5   count(*) as Users,
6   round(sum(revenue)/count(*)*100,2) as revenue
7   from online_shoppers
8   group by Session_Type
```

Session_Type	Users	revenue
Low	5596	7.65
Medium	3724	20.54
High	3010	23.75

18. Bounce Rate Segmentation

```
1 • select
2   case when BounceRates < 0.02 then 'Low'
3     when BounceRates < 0.05 then 'Medium'
4     else 'High' end as Bounce_Rate_Segments,
5   count(*) as Users,
6   round(sum(revenue)/count(*)*100,2) as conversion_rate
7   from online_shoppers
8   group by Bounce_Rate_Segments
```

Bounce_Rate_Segments	Users	conversion_rate
High	1429	1.40
Low	9633	18.70
Medium	1268	6.86

19.Best Traffic Type

```
1 • select
2   TrafficType,
3   avg(PageValues) as avg_page_values,
4   avg(ProductRelated_Duration) as avg_product_related_duration
5 from online_shoppers
6 group by TrafficType
7 order by avg_page_values desc
```

	TrafficType	avg_page_values	avg_product_related_duration
▶	20	15.128325868736614	738.3280849745779
	7	13.567345303297042	1035.0126163482666
	8	10.302435536165627	1084.488989304523
	2	8.304365890010631	1457.9416373998895
	5	7.712488764065963	742.3310294921581
	4	7.043112526850738	988.9444959520513
	10	6.2082302794191575	1258.3058496178521
	6	5.0762439099652275	1140.418002382055
	11	5.068641637259649	899.1273965796961
	14	4.936097085475922	4211.2363680326025
	9	3.818558068502517	541.7917840140207
	19	3.497520278481876	1511.367379581227
	1	3.4550741285584206	1234.0341762611206
	3	3.276033174428094	892.7577136675982
	13	2.3545854747860737	1266.75730529161
	15	0.03646820783615112	535.1455530869333

Funnel Analytics

Funnel Stage Analysis

```
1 • select sum(revenue) as purchases from online_shoppers
```

	purchases
▶	1908

Total Number of Purchases = 1908

```
1 • SELECT
2   count(*) AS Total_Visitors,
3   sum(case when ProductRelated > 10 then 1 else 0 end )as Product_Viewers,
4   sum(case when ProductRelated > 30 then 1 else 0 end) as Highly_Engaged,
5   sum(revenue) as Purchases
6 FROM online_shoppers;
```

	Total_Visitors	Product_Viewers	Highly_Engaged	Purchases
▶	12330	8157	3906	1908

Funnel Conversion %

```
1 • select
2   round(sum(case when ProductRelated > 10 then 1 else 0 end)/ count(*)*100,2) as Visitor_to_Engaged,
3   round(SUM(revenue)/ sum(case when ProductRelated > 10 then 1 else 0 end )*100,2
4   ) as Engaged_to_Purchase
5   FROM online_shoppers;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Visitor_to_Engaged	Engaged_to_Purchase		
66.16	23.39		

Month-over-Month Conversion Change

```
1 • with monthly_stats as (
2   select
3   month,
4   round(sum(revenue)/count(*)*100,2) as conversion_rate
5   from online_shoppers
6   group by month
7 )
8 select
9   month,
10  conversion_rate,
11  lag(conversion_rate) over (order by month) as previous_month,
12  conversion_rate-lag(conversion_rate) over(order by month) as difference
13  from monthly_stats
```

month	conversion_rate	previous_month	difference
Aug	17.55	NULL	NULL
Dec	12.51	17.55	-5.04
Feb	1.63	12.51	-10.88
Jul	15.28	1.63	13.65
June	10.07	15.28	-5.21
Mar	10.07	10.07	0.00
May	10.85	10.07	0.78
Nov	25.35	10.85	14.50
Oct	20.95	25.35	-4.40
Sep	19.20	20.95	-1.75

For A/B Testing we are targeting:

WEEKEND factor where:

WEEKEND = 0 = False → Weekdays

WEEKEND = 1 = True → Weekend

To Test the hypothesis that:

The purchases are done more on the weekends than the weekdays

Null Hypothesis: $\{H_0\} = (\text{Conversion Rate})_{\text{weekend}} = (\text{Conversion Rate})_{\text{weekday}}$

i.e.: There is no difference between the Conversion between weekend and weekday

Alternate Hypothesis: $\{H_1\} \rightarrow (\text{Purchase})_{\text{weekend}} > (\text{Purchase})_{\text{weekday}}$

i.e.: There is a significant difference between the conversion due to purchases carried out on the weekends than the weekdays

```
: #a/b testing for Weekend
result = df.groupby('Weekend').agg(
    visitors = ('Revenue', 'count'),
    purchases = ('Revenue', 'sum')
).reset_index()
result['conversion_rate'] = result['purchases']/result['visitors']*100
print(result)
```

	Weekend	visitors	purchases	conversion_rate
0	False	9462	1409	14.891144
1	True	2868	499	17.398884

```
visits = result['visitors']
purchase = result['purchases']
z_test, p_value = proportions_ztest(purchase, visits)
print("The z_test value is: ", z_test)
print("The p-value is : ", p_value)
if p_value<0.05:
    print("The Alternate Hypothesis is Statistically Correct")
else:
    print("The Null Hypothesis is Statistically Correct")
```

The z_test value is: -3.2529732781407104

The p-value is : 0.0011420423637110938

The Alternate Hypothesis is Statistically Correct

Similarly, for:

⇒ VisitorType:

Null Hypothesis: $\{H_0\} \rightarrow (\text{Conversion Rate})_{\text{NewVisitors}} = (\text{Conversion Rate})_{\text{ReturningVisitors}}$

i.e.: Conversion of New Visitors is same as Returning Visitors

Alternate Hypothesis: $\{H_1\} \rightarrow (\text{Purchase})_{\text{NewVisitors}} > (\text{Purchase})_{\text{ReturningVisitors}}$

i.e.: Conversion of New Visitors is more than Returning Visitors

```
# a/b testing for VisitorType : 'NewVisitor' vs 'Returning Visitor'
result = df_ab.groupby("VisitorType").agg(
    visitors = ("Revenue", 'count'),
    purchases = ("Revenue", "sum")
).reset_index()
result['conversion_rate'] = result['purchases']/result['visitors']*100
print(result)
```

	VisitorType	visitors	purchases	conversion_rate
0	New_Visitor	1694	422	24.911452
1	Returning_Visitor	10551	1470	13.932329

```
visits = result['visitors']
purchase = result['purchases']
z_test, p_value = proportions_ztest(purchase, visits)
print("The z_test value is: ", z_test)
print("The p-value is : ", p_value)
if p_value<0.05:
    print("The Alternate Hypothesis is Statistically Correct")
else:
    print("The Null Hypothesis is Statistically Correct")
```

The z_test value is: 11.605314290218352

The p-value is : 3.8726217006968784e-31

The Alternate Hypothesis is Statistically Correct

For High vs Low Product Related(Pages):

Null Hypothesis: $\{H_0\} \rightarrow (\text{Conversion Rate})_{\text{High}} = (\text{Conversion Rate})_{\text{low}}$

i.e.: Conversion due to high Product Related is same as Low

Product related pages

Alternate Hypothesis: $\{H_1\} \rightarrow (Conversion Rate)_{High} > (Conversion Rate)_{Low}$

i.e.: due to high Product Related is more than as Low Product related pages

```
#a/b testing for ProductRelated pages
#segmenting the productRelated pages into HIGH and LOW
# We are using median because it will help to segment the data into two group based on the median value found
product_related = df['ProductRelated'].median()
df['product_group'] = np.where(
    df['ProductRelated'] >= product_related,
    'High',
    'Low'
)
```

```
result = df.groupby('product_group').agg(
    visitors = ('Revenue', 'count'),
    purchases = ('Revenue', 'sum')
).reset_index()
result['conversion_rate'] = result['purchases']/result["visitors"]*100
print(result)
```

	product_group	visitors	purchases	conversion_rate
0	High	6240	1339	21.458333
1	Low	6090	569	9.343186

```
visits = result['visitors']
purchase = result['purchases']
z_test, p_value = proportions_ztest(purchase, visits)
print("The z_test value is: ", z_test)
print("The p-value is : ", p_value)
if p_value<0.05:
    print("The Alternate Hypothesis is Statistically Correct")
else:
    print("The Null Hypothesis is Statistically Correct")
```

The z_test value is: 18.597153510726894

The p-value is : 3.3882790612577407e-77

The Alternate Hypothesis is Statistically Correct

High vs Low ProductRelatedDuration

Null Hypothesis: $\{H_0\} \rightarrow (\text{Conversion Rate})_{\text{High}} = (\text{Conversion Rate})_{\text{Low}}$

i.e.: Purchases/ Conversion due to high Product Related Duration is same as Low Product Related Duration of Product related pages

Alternate Hypothesis: $\{H_1\} \rightarrow (\text{Conversion Rate})_{\text{High}} > (\text{Conversion Rate})_{\text{Low}}$

i.e.: Purchases/ Conversion due to high Product Related Duration is more than as Low Product related Duration

```
#a/b testing for Product Related Duration
# Creating 'High' and 'Low' categories for Product Related Duration
median_duration = df["ProductRelated_Duration"].median()
df['duration_groups'] = np.where(
    df['ProductRelated_Duration'] >= median_duration,
    'High',
    'Low'
)
result = df.groupby('duration_groups').agg(
    visitors = ('Revenue', 'count'),
    purchases = ('Revenue', 'sum')
).reset_index()
result['conversion_rate'] = result['purchases']/result["visitors"]*100
print(result)
```

	duration_groups	visitors	purchases	conversion_rate
0	High	6165	1374	22.287105
1	Low	6165	534	8.661800

```
visits = result['visitors']
purchase = result['purchases']
z_test, p_value = proportions_ztest(purchase, visits)
print("The z_test value is: ", z_test)
print("The p-value is : ", p_value)
if p_value<0.05:
    print("The Alternate Hypothesis is Statistically Correct")
else:
    print("The Null Hypothesis is Statistically Correct")
```

```
The z_test value is: 20.916841464509773
The p-value is : 3.7622112071127655e-97
The Alternate Hypothesis is Statistically Correct
```

High vs Low Bounce Rate

Null Hypothesis: $\{H_0\} \rightarrow (\text{Conversion Rate})_{\text{High}} = (\text{Conversion Rate})_{\text{Low}}$

i.e.: Purchases/ Conversion due to high Bounce Rate is same as Low Bounce rate

Alternate Hypothesis: $\{H_1\} \rightarrow (\text{Conversion Rate})_{\text{High}} < (\text{Conversion Rate})_{\text{Low}}$

i.e.: Purchases/ Conversion due to high Bounce Rate is less than Bounce Rate

```
# a/b testing for bounce rates
# creating 'High' and 'Low' categories for Bounce Rates
median_bounce = df['BounceRates'].median()
df['bounce_groups'] = np.where(
    df['BounceRates'] >= median_bounce,
    'High',
    'Low'
)
result = df.groupby('bounce_groups').agg(
    visitors = ('Revenue', 'count'),
    purchases = ('Revenue', 'sum')
).reset_index()
result['conversion_rate'] = result['purchases']/result["visitors"]*100
print(result)
```

	bounce_groups	visitors	purchases	conversion_rate
0	High	6165	704	11.419303
1	Low	6165	1204	19.529603

```
visits = result['visitors']
purchase = result['purchases']
z_test, p_value = proportions_ztest(purchase, visits)
print("The z_test value is: ", z_test)
print("The p-value is : ", p_value)
if p_value<0.05:
    print("The Alternate Hypothesis is Statistically Correct")
else:
    print("The Null Hypothesis is Statistically Correct")
```

```
The z_test value is: -12.450500871732007
The p-value is : 1.3897028573513229e-35
The Alternate Hypothesis is Statistically Correct
```